

Docker-02

1. Create a tomcat container on 8080 and deploy a sample application in tomcat.

Pull Tomcat Image

```
ubuntu@ip-172-31-10-98:~$ docker pull tomcat\
>
Using default tag: latest
latest: Pulling from library/tomcat
97c47a5c1a2f: Pull complete
4f4fb700ef54: Pull complete
b38893372039: Pull complete
d92a694d8aa0: Pull complete
6b82a07ff6e7: Pull complete
65313903a8a8: Pull complete
b40150c1c271: Pull complete
f80ef884f8c2: Download complete
13883d0clf97: Download complete
Digest: sha256:247650787b97751d68aa2d532d6e09111c36946d9a50ea0e5071e19facc1e5e7
Status: Downloaded newer image for tomcat:latest
docker.io/library/tomcat:latest
```

Run Tomcat Container : `docker run -d --name tomcat-app -p 8080:8080 tomcat`

```
ubuntu@ip-172-31-10-98:~$ docker run -d --name tomcat-app -p 8080:8080 tomcat
3b86ee2835bebb12db9b88bba13a4b20ee553246cfdeb2b7222a99cd1a8d89d5
```

Download Sample WAR File : `wget`

`https://tomcat.apache.org/tomcat-7.0-doc/appdev/sample/sample.war`

```
ubuntu@ip-172-31-10-98:~$ wget https://tomcat.apache.org/tomcat-7.0-doc/appdev/sample/sample.war
--2026-06-01 16:55:34-- https://tomcat.apache.org/tomcat-7.0-doc/appdev/sample/sample.war
Resolving tomcat.apache.org (tomcat.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to tomcat.apache.org (tomcat.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 4606 (4.5K)
Saving to: 'sample.war'

sample.war                                     100%[=====]
2026-06-01 16:55:35 (78.7 MB/s) - 'sample.war' saved [4606/4606]
```

Copy WAR File into Tomcat : `docker cp sample.war`

`tomcat-app:/usr/local/tomcat/webapps/`

```
ubuntu@ip-172-31-10-98:~$ docker cp sample.war tomcat-app:/usr/local/tomcat/webapps/
Successfully copied 6.14kB to tomcat-app:/usr/local/tomcat/webapps/
ubuntu@ip-172-31-10-98:~$
```

```
i.08ba2672d4332489d (docker test)
```

Verify Application



2. Create volume and deploy tomcat container on port 8081.

Create Docker Volume :Create Docker Volume

```
ubuntu@ip-172-31-10-98:~$ docker volume ls
DRIVER      VOLUME NAME
local       309262feab9e0976f5e4446ee95646c0bf32b61f8da48641a298b129c0b78d8c
local       tomcat-volume
```

Run Tomcat with Volume :docker run -d \
--name tomcat-volume-app \
-p 8081:8080 \
-v tomcat-volume:/usr/local/tomcat/webapps \
tomcat

```
root@ip-172-31-10-98:~# cd /var/lib/docker/volumes/
root@ip-172-31-10-98:/var/lib/docker/volumes# ls
309262feab9e0976f5e4446ee95646c0bf32b61f8da48641a298b129c0b78d8c  backingFsBlockDev  metadata.db  tomcat-volume
root@ip-172-31-10-98:/var/lib/docker/volumes# cd tomcat-volume/
root@ip-172-31-10-98:/var/lib/docker/volumes/tomcat-volume# ls
data
root@ip-172-31-10-98:/var/lib/docker/volumes/tomcat-volume# cd data/
```

Cd /var/lib/docker/volumes/tomcat-volume/_data

Download Sample WAR File : wget

<https://tomcat.apache.org/tomcat-7.0-doc/appdev/sample/sample.war>

```
root@ip-172-31-10-98:/var/lib/docker/volumes/tomcat-volume# cd _data/
root@ip-172-31-10-98:/var/lib/docker/volumes/tomcat-volume/_data# ls
root@ip-172-31-10-98:/var/lib/docker/volumes/tomcat-volume/_data# wget https://tomcat.apache.org/tomcat-6.0-doc/appdev/sample/sample.war
--2026-06-02 06:41:09-- https://tomcat.apache.org/tomcat-6.0-doc/appdev/sample/sample.war
Resolving tomcat.apache.org (tomcat.apache.org)... 151.101.2.132, 2a04:4e42:644
Connecting to tomcat.apache.org (tomcat.apache.org)[151.101.2.132]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 4606 (4.5K)
Saving to: 'sample.war'

sample.war          100%[=====>]  4.50K  --.-KB/s  in 0s

2026-06-02 06:41:09 (82.7 MB/s) - 'sample.war' saved [4606/4606]

root@ip-172-31-10-98:/var/lib/docker/volumes/tomcat-volume/_data# ls
```

Create tomcat container on 8081 port number and run the container

docker container run -itd -p 8081:8080 -v

/var/lib/docker/volumes/tomcat-volume/_data:/usr/local/tomcat/webapps

tomcat:latest

```
root@ip-172-31-10-98:/var/lib/docker/volumes/tomcat-volume/_data# docker container run -itd -p 8081:8080 -v /var/lib/docker/volumes/tomcat-volume/_data:/usr/local/tomcat/webapps tomcat:latest
4f3290f8147b770a23598d2fc8fb1580c59faa189139eae653d217a06ce5af2b
```

Run container

```
root@ip-172-31-10-98:/var/lib/docker/volumes/tomcat-volume/_data# docker exec -it 4f3290f8147b770a23598d2fc8fb1580c59faa189139eae653d217a06ce5af2b /bin/bash
root@4f3290f8147b:/usr/local/tomcat# ls
bin      conf     filtered-KEYS  LICENSE  native-jni-lib  README.md  RUNNING.txt  upstream-KEYS  webapps.dist
BUILDING.txt  CONTRIBUTING.md  lib          NOTICE  RELEASE-NOTES   webapps
root@4f3290f8147b:/usr/local/tomcat# cd webapps
root@4f3290f8147b:/usr/local/tomcat/webapps# ls
bash: ls: command not found
root@4f3290f8147b:/usr/local/tomcat/webapps# ls
sample  sample.war
root@4f3290f8147b:/usr/local/tomcat/webapps# ll
total 24
drwxr-xr-x 3 root root 4096 Jun  2 06:44 ./
drwxr-xr-x 1 root root 4096 May  5 23:13 ../
drwxr-xr-x 5 root root 4096 Jun  2 06:44 sample/
-rw-r--r-- 1 root root 4606 Nov 16 2016 sample.war
```

Verify application



Sample "Hello, World" Application

This is the home page for a sample application used to illustrate the source directory organization of a web application utilizing the principles outlined in the Application Developer's Guide.

To prove that they work, you can execute either of the following links:

- To a [JSP page](#).
- To a [servlet](#).

3. Limit the nginx container to 500 MB.

Pull Nginx

```
root@ip-172-31-10-98:~# docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:5aca99593157f4ae539a5dec1092a0ad0762f0e2eb1789085a13a0f5622369f6
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
root@ip-172-31-10-98:~# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
nginx:latest	5aca99593157	241MB	66MB	0
tomcat:latest	247650787b97	589MB	158MB	0

Run Container with Memory Limit :

```
docker run -d \
--name nginx-limit \
-p 80:80 \
--memory="500m" \
nginx
```

```
root@ip-172-31-10-98:~# docker run -d \
--name nginx-limit \
-p 80:80 \
--memory="500m" \
nginx
ea7e4855cce68279b50ea84a691d7e6a4b0c3ca92f127ebd6f9877f59b933a81
root@ip-172-31-10-98:~# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
ea7e4855cce6	nginx	"/docker-entrypoint..."	17 seconds ago	Up 16 seconds	0.0.0.0:80->80/tcp, [::]:80->80/tcp	nginx-limit
4f3290f8147b	tomcat:latest	"catalina.sh run"	25 minutes ago	Up 25 minutes	0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp	romantic_gagarin

Verify : docker inspect nginx-limit | grep Memory

```
root@ip-172-31-10-98:~# docker inspect nginx-limit | grep Memory
      "Memory": 524288000,
      "MemoryReservation": 0,
      "MemorySwap": 1048576000,
      "MemorySwappiness": null,
root@ip-172-31-10-98:~#
```

4. Create a sample docker file using below instructions.

1. Base module as amazonlinux:latest
2. Maintainer you name
3. Install nginx
4. COPY one index.html file to image
5. Expose on port 80
6. Command to start the nginx container

Create Directory

```
root@ip-172-31-10-98:~# mkdir nginx-image
root@ip-172-31-10-98:~# cd nginx-image/
root@ip-172-31-10-98:~/nginx-image# vi index.html
root@ip-172-31-10-98:~/nginx-image# vi Dockerfile
```

Create index.html

```
root@ip-172-31-10-98:~/nginx-image# cat index.html
<html>
<head>
<title>Docker Demo</title>
</head>
<body>
<h1>Welcome to Docker Nginx</h1>
</body>
</html>
```

Create Dockerfile

```
root@ip-172-31-10-98:~/nginx-image# cat Dockerfile
FROM amazonlinux:latest

LABEL maintainer="Bhagav"

RUN yum update -y && \
    amazon-linux-extras install nginx1 -y || yum install nginx -y

COPY index.html /usr/share/nginx/html/index.html

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```

Build Image :docker build -t custom-nginx:v1 .

```
root@ip-172-31-10-98:~/nginx-image# docker build -t custom-nginx:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  3.072kB
Step 1/6 : FROM amazonlinux:latest
latest: Pulling from library/amazonlinux
c7345fb8c651: Pulling fs layer
c7345fb8c651: Download complete
b3d837a98232: Download complete
2da5356a79d8: Download complete
c7345fb8c651: Pull complete
Digest: sha256:ad882807c6c88f478186fa2632637380e06071a303e2b3bd96ab71846a77c6a3
Status: Downloaded newer image for amazonlinux:latest
--> ad882807c6c8
```

```
Complete!
----> Removed intermediate container cfb5a00c569d
----> 016267bbca91
Step 4/6 : COPY index.html /usr/share/nginx/html/index.html
----> 3f7304e6dd5c
Step 5/6 : EXPOSE 80
----> Running in 489d301ba612
----> Removed intermediate container 489d301ba612
----> 4dc12cc9b41c
Step 6/6 : CMD ["nginx", "-g", "daemon off;"]
----> Running in c2ecca66060e
----> Removed intermediate container c2ecca66060e
----> 20e7ecf3e63b
Successfully built 20e7ecf3e63b
Successfully tagged custom-nginx:v1
```

```
root@ip-172-31-10-98:~/nginx-image# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
amazonlinux:latest	ad882807c6c8	227MB	59.3MB	
custom-nginx:v1	20e7ecf3e63b	584MB	195MB	
nginx:latest	5aca99593157	241MB	66MB	U
tomcat:latest	247650787b97	589MB	158MB	U

Run Image :`docker run -d -p 80:80 custom-nginx:v1`

```
root@ip-172-31-10-98:~/nginx-image# docker run -d -p 80:80 custom-nginx:v1
\10ce6eba33afa6fcb0e4a937b07257086c18db48c9d1e879587cb87f3df231a5
docker: Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint c
9fd41a15a42eb075da137b4cc5236ce5eee952239de903c): Bind for 0.0.0.0:80 failed: port is already allocated

Run 'docker run --help' for more information
```

```
root@ip-172-31-10-98:~/nginx-image# docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
10ce6eba33af	custom-nginx:v1	"nginx -g 'daemon of..."	46 seconds ago	Created		crazy_matsumoto
ea7e4855cce6	nginx	"/docker-entrypoint..."	19 minutes ago	Up 19 minutes	0.0.0.0:80->80/tcp, [::]:80->80/tcp	nginx-limit
4f3290f0147b	tomcat:latest	"catalina.sh run"	44 minutes ago	Up 44 minutes	0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp	romantic_gagarin
2ca9a64a2c10	tomcat	"catalina.sh run"	14 hours ago	Exited (143) 14 hours ago		tomcat-volume-app
3b86ee2835be	tomcat	"catalina.sh run"	15 hours ago	Exited (143) 14 hours ago		tomcat-app
ec8124ebbbb3	01c92ffef29	"/usr/bin/tini -- /u..."	2 days ago	Exited (143) 2 days ago		jenkins
9d4db9bd4569	nginx	"/docker-entrypoint..."	2 days ago	Exited (0) 2 days ago		nginx-container
a80206ee4319	0e760dfbc48	"/hello"	2 days ago	Exited (0) 2 days ago		quizzical_proskuriak

```
ova
root@ip-172-31-10-98:~/nginx-image#
```

5. Push image to Docker hub

Login :`docker login`

```
root@ip-172-31-10-98:~/nginx-image# docker login
Authenticating with existing credentials... [Username: bhargav0117]

Info → To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded
```

Tag Image: `docker tag custom-nginx:v1 <dockerhub-username>/custom-nginx:v1`

Push : `docker push <dockerhub-username>/custom-nginx:v1`

```
root@ip-172-31-10-98:~/nginx-image# docker tag custom-nginx:v1 bhargav0117/custom-nginx:v1
root@ip-172-31-10-98:~/nginx-image# docker push bhargav0117/custom-nginx:v1
The push refers to repository [docker.io/bhargav0117/custom-nginx]
0388bb088a55: Pushed
c7345fb8c651: Mounted from library/amazonlinux
6fceeefb6b2e8: Pushed
v1: digest: sha256:20e7ecf3e63bd2507f29f85d7ca500ad6572b2d58acf6589dd2ec47f080e6e90 size: 972
```

Verify docker hub

The screenshot shows the Docker Hub interface for the repository `bhargav0117/custom-nginx`. The page includes a sidebar with navigation options like Repositories, Hardened Images, Collaborations, and Settings. The main content area shows the repository details, including the last push time (2 minutes ago), repository size (186.4 MB), and a table of tags. The 'Tags' table shows a single tag `v1` with a pulled time of 'less than 1 day' and a pushed time of '2 minutes'. There is also a 'Docker commands' section with a public view button and a 'buildcloud' advertisement.

Tag	OS	Type	Pulled	Pushed
v1	linux	Image	less than 1 day	2 minutes

6. push image to AWS ECR

Create Repository

The screenshot shows the Amazon ECR Private repositories page. The left sidebar contains navigation options for Amazon Elastic Container Registry, including Private registry, Public registry, and ECR public gallery. The main content area shows a list of private repositories. A table lists the repository `custom-nginx` with its URI, creation date, tag immutability, and encryption type.

Repository name	URI	Created at	Tag immutability	Encryption type
custom-nginx	922345236105.dkr.ecr.ap-south-1.amazonaws.com/custom-nginx	02 June 2026, 16:21:32 (UTC+05.5)	Mutable	AES-256

Configure aws cli:

```
root@ip-172-31-10-98:~/nginx-image# aws configure
AWS Access Key ID [None]: AKIA5NQAPM2EQX2PRMV5
AWS Secret Access Key [None]: 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
Default region name [None]: ap-south-1
Default output format [None]: json
```

Login to ECR :aws ecr get-login-password --region ap-south-1 | \
docker login --username AWS \
--password-stdin <aws-account-id>.dkr.ecr.ap-south-1.amazonaws.com

```
root@ip-172-31-10-98:~/nginx-image# aws ecr get-login-password --region ap-south-1 | docker login --username AWS --password-stdin 922345236105.dkr.ecr.ap-south-1.amazonaws.com
WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/
Login Succeeded
```

Tag Image :docker tag custom-nginx:v1 \
<aws-account-id>.dkr.ecr.ap-south-1.amazonaws.com/custom-nginx:v1

Push Image :docker push \
<aws-account-id>.dkr.ecr.ap-south-1.amazonaws.com/custom-nginx:v1

```
root@ip-172-31-10-98:~/nginx-image# docker tag custom-nginx:v1 \
922345236105.dkr.ecr.ap-south-1.amazonaws.com/custom-nginx:v1
root@ip-172-31-10-98:~/nginx-image# docker push \
922345236105.dkr.ecr.ap-south-1.amazonaws.com/custom-nginx:v1
The push refers to repository [922345236105.dkr.ecr.ap-south-1.amazonaws.com/custom-nginx]
0388bb088a55: Pushed
c7345fb8c651: Pushed
6fcee6b6b2e8: Pushed
v1: digest: sha256:20e7ecf3e63bd2507f29f85d7ca500ad6572b2d58acf6589dd2ec47f080e6e90 size: 972
root@ip-172-31-10-98:~/nginx-image#
```


Veriry aws console

Amazon ECR > Private registry > Repositories > custom-nginx

Amazon Elastic Container Registry

▼ Private registry

Repositories

custom-nginx

Features & Settings

▼ Public registry

Repositories

Settings

ECR public gallery

Amazon ECS

Amazon EKS

custom-nginx

SummaryImagesLifecycle policyPermissionsRepository tags

Images (1) Info

Filter active images

	Image tags	Type	Created at	Image size (MB)	Image digest	Last pulled at
	v1	Image	02 June 2026, 16:41:17 (UTC+05.5)	195.48	sha256:20e7ecf...	-